



Wings for Science in Archipelagos



Archipelagos, Institute of Marine Conservation is a Greek non-profit, non-governmental organization based in the Aegean Sea. During the last 15 years, the institute has been committed to researching and defending the biodiversity of the Greek seas and its islands. Work focuses on a combination of multi-disciplinary scientific research with efficient conservation work, in which the local communities share an active part.

During a week's stay on Samos island, Wings for Science worked in collaboration with Archipelagos Institute. The Archipelagos team utilised the institutes' 40ft sailing boat, Nireas, to survey the sea, whilst Wings for Science simultaneously undertook aerial observations, focusing on the following:

1. Mapping of *Posidonia oceanica* seagrass meadows
2. Surveying marine mammals, including dolphins and sperm whales
3. Monitoring islet erosion, with the production of 3D models



Joint Aerial & Boat-based Surveys for Marine Conservation

Marine Mammal Surveys

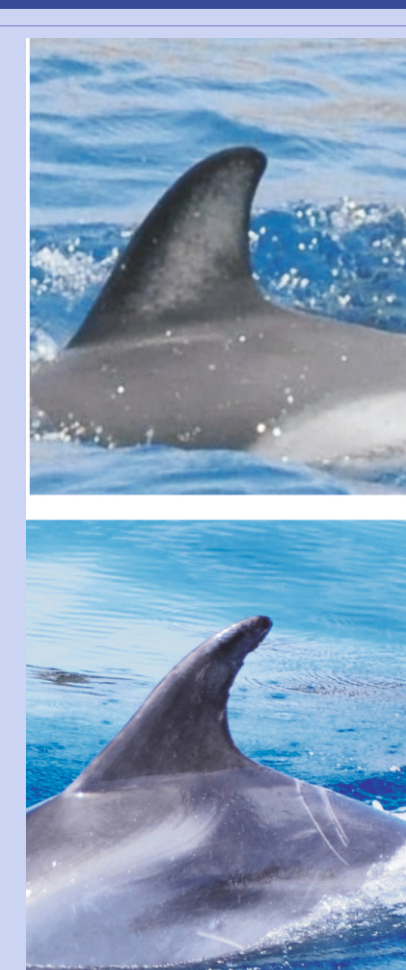
The combined efforts of the Nireas boat and the Wings for Science aeroplane enabled unparalleled observation from both sea and sky. This particular methodology of joint survey technique enabled sightings of pods of both Common dolphins and Bottlenose dolphins and the extraordinary sighting of three sperm whales.



Sperm Whales

Sperm whales (*Physeter macrocephalus*) migrate through the Aegean Sea and are in grave danger from a number of increasing anthropogenic threats, such as vessel collisions, overfishing, pollution and military activity (sonar, explosions). More data on their migration routes and behaviour is necessary to better understand the threats and develop efficient conservation strategies.

During one joint aerial and boat based survey, sperm whales were not located by either hydrophone or aerial observation. However, on the departure of Wings for Science, the two pilots recorded in the same region a small social group, comprising of two females and a calf. This proved that aerial surveys over a vast marine area can become an invaluable tool to locate pods of large marine mammals such as sperm whales, which the research boats can later follow for closer behavioural and acoustic observations.



Dolphin Fin Identification

The Common dolphin (*Delphinus delphis*) and the Bottlenose dolphin (*Tursiops truncatus*) are two of the four dolphin species inhabiting the Greek seas. Due to overfishing, pollution, trawlers, and other threats, these species are in danger and therefore effective conservation is essential for their protection. Common dolphins were once one of the most common cetacean species in the Mediterranean Sea, however during the last few decades they have declined throughout the region. Pods were sighted by aerial surveys, providing coordinates for Nireas sailboat to locate individuals and take detailed photographs for later photo-identification. This technique helps to obtain information on group structure, site fidelity, movement patterns and population size.

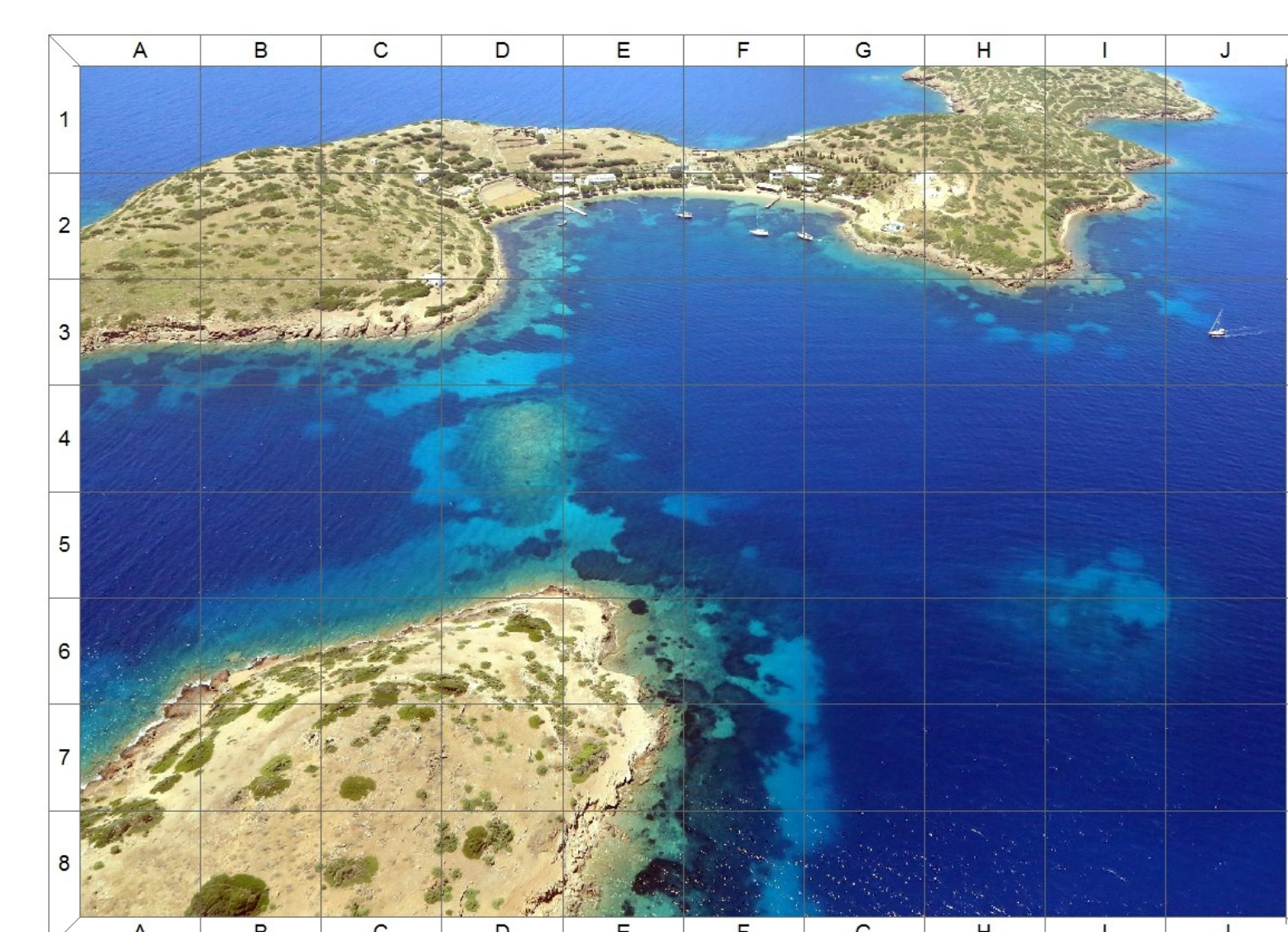


Mapping of Posidonia Seagrass Meadows

Posidonia oceanica is a seagrass endemic to the Mediterranean, playing a fundamental role in the health and productivity of marine ecosystems and fisheries. Due to an extremely slow growth rate, the seagrass is considered the oldest living organism on Earth – dated as over 100, 000 years old.

Unfortunately, recent scientific research has shown that *P. oceanica* is declining faster than the rate at which it spreads. Due to anthropogenic factors, including coastal construction, pollution, trawling and fish farms, the seagrass is declining at approximately 5% every year.

Although 25,000-45,000 km² of *P. oceanica* meadows have been recorded in the Mediterranean Basin, there is a significant lack of mapping, especially in the Aegean, which hinders our ability to protect this habitat. The oblique aerial photography conducted by Wings for Science provides data for extensive mapping, which allows Archipelagos Institute to assess the extent of *P. oceanica* in shallow waters. Using this essential primary data, Archipelagos Institute can conduct ground truthing using sonar scanning, an underwater HD camera and grab sampling. These joint surveys provide invaluable information regarding the health and density of seagrass meadows, which will help reverse damage to the forests of our seas.



Soil Erosion of Islet Ecosystems

In the small islets of the Aegean Sea, soil erosion is a major threat to the biodiversity of terrestrial ecosystems. Wings for Science took successive aerial photographs of small islets, which were processed with the geoscale software application (www.geoscale.org). The end product was a 3D model of each individual islet, which will then be analyzed for the assessment of erosion rates. This will be done by the comparison of the attained 3D model and either prior or future photographic material of the specific islets.

